

Model Selection &

DISTRIBUTION MODEL SELECTION

Test	Comparison	Statistic
Likelihood Ratio	Normal vs Student-t	310.7531
Vuong	Student-t vs GED	-3.3283

VOLATILITY MODEL SELECTION

Test	Comparison	Statistic
Likelihood Ratio	GARCH vs GJR-GARCH	4.5200
Ljung-Box (sq resid)	GARCH adequacy (lag 10)	26.7556
ARCH-LM	Remaining ARCH effects	27.1003

INTERPRETATION

The LR test decisively rejects the Normal distribution ($p < 0.001$) in favor of Student-t, confirming heavy tails.

The Vuong test identifies GED as statistically superior to Student-t, though both capture tails far better than Normal.

GJR-GARCH is preferred over GARCH(1,1) ($p = 0.034$), confirming a significant leverage (asymmetry) effect.

GARCH residual diagnostics FAIL: remaining ARCH effects suggest a higher-order or regime-switching specification may be needed.

This failure is an honest finding -- the GARCH(1,1) captures most but not all volatility dynamics in this asset.

Diagnostic Tests

p-value	Result
0.0000	Student-t preferred
0.0009	GED preferred

p-value	Result
0.0335	GARCH preferred (leverage effect significant)
0.0028	FAIL
0.0025	FAIL

may be needed.

Out-of-

Training: 1277 obs |

VOLATILITY FORECAST ACCURACY

Metric	GARCH(1,1)	GJR-GARCH
RMSE vs Realized	1.0834	1.0795
MZ Alpha (ideal: 0)	-0.7186	-
MZ Beta (ideal: 1)	1.0653	-
MZ R-squared	0.3401	-
MZ Joint Test p-value	0.0000	-
DM Test p-value	0.6793	-

OOS VaR BACKTEST RESULTS

Method & Level	Violations	Rate
Historical 95	19	3.47%
Normal 95	17	3.10%
Student-t 95	19	3.47%
Historical 99	2	0.36%
Normal 99	5	0.91%
Student-t 99	2	0.36%

Sample Validation Results

(to 2024-12-09) | Test: 548 obs (from 2024-12-10)

Winner
GJR-GARCH
-
-
-
FAIL

Significant difference between models

Expected	Kupiec	Christoffersen
5.00%	PASS (p=0.082)	PASS (p=0.242)
5.00%	FAIL (p=0.029)	PASS (p=0.296)
5.00%	PASS (p=0.082)	PASS (p=0.242)
1.00%	PASS (p=0.086)	PASS (p=0.904)
1.00%	PASS (p=0.834)	PASS (p=0.761)
1.00%	PASS (p=0.086)	PASS (p=0.904)

Cond. Coverage
PASS (p=0.111)
PASS (p=0.053)
PASS (p=0.111)
PASS (p=0.227)
PASS (p=0.934)
PASS (p=0.227)

Bootstrap Confidence Intervals (500 iterations)

GARCH PARAMETER UNCERTAINTY

Parameter	Mean	95% CI Lower
GARCH Alpha	0.1354	0.0626
GARCH Beta	0.7478	0.3447
Persistence (Alpha+Beta)	0.8833	0.5720
99% Historical VaR	-0.0778	-0.0962
GPD Shape (Xi)	0.0466	-0.1355

TAIL ESTIMATION UNCERTAINTY

Effective tail sample: 92.0 exceedances

GPD xi CI width: 0.3680 (wide interval = high uncertainty)

The sign of xi is uncertain: CI spans negative to positive, meaning we cannot confidently determine whether the tail is heavy ($\xi > 0$) or thin ($\xi < 0$) from this sample alone.

EVT THRESHOLD SENSITIVITY

Threshold (%ile)	Exceedances	GPD Xi
1th	19	-0.1125
2th	37	0.0741
3th	55	0.1080
5th	92	0.0624
7th	128	0.0824
10th	183	0.1372

tions, block=20 days)

95% CI Upper	Std Error
0.2682	0.0505
0.8843	0.1280
0.9798	0.0980
-0.0654	0.0078
0.2325	0.0988

Xi Std Error	EVT VaR 99%
0.2036	-8.18%
0.1766	-8.03%
0.1494	-8.01%
0.1108	-8.12%
0.0957	-8.08%
0.0841	-7.98%

Rolling 2-Year Parameter Stability

Date	GARCH Alpha	GARCH Beta	Persistence
2022-10-29	0.0000	0.9980	0.9980
2022-11-18	0.0358	0.8662	0.9020
2022-12-08	0.0470	0.8294	0.8764
2022-12-28	0.0638	0.8756	0.9394
2023-01-17	0.1058	0.8854	0.9912
2023-02-06	0.0863	0.8923	0.9787
2023-02-26	0.1011	0.8579	0.9590
2023-03-18	0.1571	0.8366	0.9937
2023-04-07	0.1351	0.8230	0.9581
2023-04-27	0.1499	0.8056	0.9555
2023-05-17	0.1568	0.8082	0.9650
2023-06-06	0.1665	0.8294	0.9960
2023-06-26	0.1730	0.8070	0.9799
2023-07-16	0.1611	0.8013	0.9624
2023-08-05	0.1954	0.8046	1.0000
2023-08-25	0.1681	0.8319	1.0000
2023-09-14	0.1154	0.8846	1.0000
2023-10-04	0.1309	0.8691	1.0000
2023-10-24	0.1196	0.8804	1.0000
2023-11-13	0.8302	0.1698	1.0000
2023-12-03	0.8772	0.1228	1.0000
2023-12-23	0.8303	0.1697	1.0000
2024-01-12	0.8627	0.1373	1.0000
2024-02-01	0.7275	0.1116	0.8391
2024-02-21	0.8757	0.1021	0.9779
2024-03-12	0.9214	0.0786	1.0000
2024-04-01	0.9089	0.0911	1.0000
2024-04-21	0.0616	0.9384	1.0000
2024-05-11	0.0671	0.9329	1.0000
2024-05-31	0.5177	0.1359	0.6536
2024-06-20	0.0543	0.9457	1.0000
2024-07-10	0.0476	0.9524	1.0000
2024-07-30	0.0385	0.9615	1.0000
2024-08-19	0.0527	0.9473	1.0000
2024-09-08	0.0386	0.9614	1.0000
2024-09-28	0.0401	0.9599	1.0000
2024-10-18	0.0455	0.9545	1.0000
2024-11-07	0.0121	0.9879	1.0000
2024-11-27	0.0047	0.9953	1.0000
2024-12-17	0.0103	0.9897	1.0000
2025-01-06	0.0186	0.9814	1.0000
2025-01-26	0.0126	0.9786	0.9912
2025-02-15	0.0202	0.9567	0.9769
2025-03-07	0.0856	0.6056	0.6912

2025-03-27	0.0570	0.8405	0.8976
2025-04-16	0.0656	0.8191	0.8847
2025-05-06	0.0759	0.8006	0.8765
2025-05-26	0.0736	0.8113	0.8849
2025-06-15	0.0728	0.8522	0.9250
2025-07-05	0.0532	0.9140	0.9673
2025-07-25	0.0409	0.9369	0.9778
2025-08-14	0.0469	0.9289	0.9758
2025-09-03	0.0474	0.9144	0.9619
2025-09-23	0.0505	0.9286	0.9791
2025-10-13	0.0584	0.9130	0.9714
2025-11-02	0.0507	0.9258	0.9766
2025-11-22	0.0460	0.9321	0.9781
2025-12-12	0.0415	0.9352	0.9767
2026-01-01	0.0527	0.9164	0.9691
2026-01-21	0.0568	0.9062	0.9630
2026-02-10	0.0847	0.8792	0.9639
2026-03-02	0.0896	0.8473	0.9370
2026-03-22	0.0702	0.8934	0.9637
2026-04-11	0.0795	0.8694	0.9489
2026-05-01	0.0931	0.8388	0.9318
2026-05-21	0.0913	0.8470	0.9383
2026-06-11	0.1255	0.7664	0.8919

GPD Xi
-0.0518
-0.0323
-0.0323
-0.0323
-0.0323
0.0787
0.2148
0.1718
0.1619
0.0303
0.0066
0.0093
0.1055
0.1212
0.0165
-0.0312
-0.0005
0.0904
0.1581
0.3359
0.1777
0.1727
0.1680
0.2131
0.2201
0.1496
-0.3745
-0.3614
-0.3674
-0.3674
-0.3665
-0.3963
-0.3074
-0.3335
-0.3068
-0.3381
-0.3044
-0.2890
-0.3044
-0.3044
-0.2651
-0.2671
-0.2671
-0.2803

-0.2927
-0.4222
-0.4201
-0.4132
-0.2591
-0.2591
-0.1686
-0.1561
-0.0294
0.0237
-0.0472
-0.0472
-0.1218
-0.1395
0.0223
0.0601
0.2817
0.2426
0.2468
0.2457
0.2554
0.3185
0.2953

KEY RISK METRICS

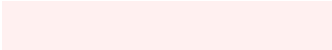
Metric	Value	Annualized
Daily Volatility	2.81%	44.63%
Skewness		-0.2145
Excess Kurtosis		3.9708
VaR 95% (Historical)	-4.31%	
VaR 99% (Historical)	-8.05%	
CVaR 99% (Historical)	-10.80%	
Maximum Drawdown	-76.63%	
GARCH Persistence		0.9961
Volatility Half-Life		175.16
Jarque-Bera p-value		0.0000

OVERALL RISK ASSESSMENT: VERY HIGH

PROJECT APEX - Quantitative Risk Dashboard

Analysis Period: 2021-06-12 to 2026-06-11 | 1825 observations

Interpretation
HIGH - typical for digital assets
Negative - left tail risk
Heavy tails - extreme events more likely
\$43,137 on \$1M
\$80,545 on \$1M
\$108,043 on \$1M
On 2022-11-21
Near-IGARCH: shocks decay very slowly
175.16227496502933 days for shock to halve
REJECT normality (p < 0.001)



Value at Risk & Expected Shortfall

Portfolio Value: \$1,000,000 | All figures are

VALUE AT RISK (VaR)

Method	VaR 90%	\$ Loss 90%	VaR 95%
Historical	-3.04%	\$30,355	-4.31%
Parametric (Normal)	-3.57%	\$35,749	-4.60%
Parametric (Student-t)	-3.04%	\$30,437	-4.46%
GARCH(1,1)	-3.57%	\$35,704	-4.59%
Monte Carlo (100k)	-3.57%	\$35,743	-4.59%

EXPECTED SHORTFALL (CVaR)

Method	CVaR 90%	\$ Loss 90%	CVaR 95%
Historical	-5.15%	\$51,458	-6.70%
Parametric (Normal)	-4.91%	\$49,058	-5.77%
Monte Carlo (100k)	-4.90%	\$49,044	-5.78%

EXTREME VALUE THEORY (EVT) - Peaks Over Threshold

Confidence	EVT VaR	EVT ES	GPD Shape (xi)
95.0%	-4.33%	-1.95%	0.0624
99.0%	-8.12%	-5.99%	
99.9%	-14.24%	-12.51%	

tfall Analysis

1-day horizon

\$ Loss 95%	VaR 99%	\$ Loss 99%
\$43,137	-8.05%	\$80,545
\$45,962	-6.51%	\$65,120
\$44,620	-9.26%	\$92,583
\$45,913	-6.51%	\$65,065
\$45,924	-6.54%	\$65,354

\$ Loss 95%	CVaR 99%	\$ Loss 99%
\$66,960	-10.80%	\$108,043
\$57,709	-7.46%	\$74,646
\$57,768	-7.48%	\$74,796

Tail Index
16.01

Volatility Model Parameters

GARCH(1,1) with Student-t Innovations

Parameter	Value	Description
omega	0.1323	Constant variance component
alpha (ARCH)	0.0601	Shock impact coefficient
beta (GARCH)	0.9360	Persistence coefficient
alpha + beta	0.9961	Total persistence (near 1 = high)
Long-run vol (daily %)		Unconditional daily volatility
Half-life (days)	175.1623	Days for shock to decay 50%

GJR-GARCH(1,1,1) - Leverage Effect

Parameter	Value	Description
gamma (leverage)	0.0469	Asymmetric response to negative
alpha	0.0444	Symmetric shock impact
beta	0.9245	Persistence
Leverage effect?		YES - negative returns increase vc

10-Day Volatility Forecast (GARCH)

Day	Daily Vol (%)	Annualized Vol (%)
T+1	2.7457	43.59
T+2	2.7644	43.88
T+3	2.7828	44.17
T+4	2.8010	44.46
T+5	2.8190	44.75
T+6	2.8368	45.03
T+7	2.8545	45.31
T+8	2.8720	45.59
T+9	2.8893	45.87
T+10	2.9064	46.14

Parameters & Forecasts

shocks

more than positive

Stress Testing & Scenario Analysis

HYPOTHETICAL STRESS SCENARIOS (\$1M Portfolio)

Scenario	Shock (%)	Dollar Loss
Moderate correction (-15%)	-15.00%	\$150,000
Severe crash (-30%)	-30.00%	\$300,000
Black swan (-50%)	-50.00%	\$500,000
Historical worst month	-47.43%	\$474,314
Historical worst week	-33.04%	\$330,373
3-sigma daily event	-8.41%	\$84,058
5-sigma daily event	-14.03%	\$140,282

VOLATILITY REGIME DISTRIBUTION

Regime	Days	Percentage
Low vol (<40% ann)	906	50.45%
Medium vol (40-70% ann)	846	47.10%
High vol (70-100% ann)	44	2.45%
Crisis vol (>100% ann)	0	0.00%

alysis

Severity
HIGH
SEVERE
EXTREME
EXTREME
EXTREME
HIGH
HIGH

30-Day Price Simulation Res

GEOMETRIC BROWNIAN MOTION (10,000 paths)

Metric	Value
Current Price	\$61,976
Mean 30-Day Price	\$62,515
Median 30-Day Price	\$61,957
Std Dev of Final Price	\$9,756
5th Percentile	\$47,689
25th Percentile	\$55,648
75th Percentile	\$68,435
95th Percentile	\$79,508
P(Loss)	50.08%
P(>10% Gain)	25.74%
P(>20% Loss)	8.15%
Min Simulated	\$36,205
Max Simulated	\$115,532

MERTON JUMP-DIFFUSION MODEL

Metric	Value
Mean 30-Day Price	\$60,065
Median 30-Day Price	\$59,371
5th Percentile	\$44,383
95th Percentile	\$77,883
P(Loss)	59.81%
P(>20% Loss)	15.03%

ults

Return Distribution Fit

DISTRIBUTION FIT COMPARISON

Distribution	AIC	BIC
Normal (Gaussian)	-7854.08	-7843.06
Student-t	-8162.83	-8146.31
Skewed-t (NCT)	-8168.75	-8146.71
Generalized Error (GED)	-8202.29	-8185.76

NORMALITY TESTS

Test	Statistic	p-value
Jarque-Bera	1204.3968	0.000000
Shapiro-Wilk	0.9470	0.000000
Anderson-Darling	23.7325	

DESCRIPTIVE STATISTICS

Statistic	Value
Observations	1825
Mean (daily)	0.000278
Mean (annualized)	0.069985
Std Dev (daily)	0.028112
Std Dev (annualized)	0.446262
Skewness	-0.214538
Excess Kurtosis	3.970776
Minimum Return	-0.174053
Maximum Return	0.135764
1st Percentile	-0.080545
5th Percentile	-0.043137
Median	-0.000104
95th Percentile	0.045127
99th Percentile	0.079276

ting & Normality Tests

Best Fit?
YES

Result
REJECT
REJECT
REJECT (stat >> critical values)

VaR Model Backtesting Results

KUPIEC PROPORTION OF FAILURES TEST

Metric	Value
Test Window	1573 observations
VaR Level	99%
Actual Violations	15
Expected Violations	16
Violation Rate	0.95%
Kupiec LR Statistic	0.0348
Kupiec p-value	0.8521
Result	PASS

